

Empirical UX Evaluation: Data Collection

Highlights

- Empirical ways of generating and collecting data within the needs pyramid
- Empirical methods and techniques for generating and collecting qualitative UX data
 - Critical incident identification
 - User think-aloud technique
 - Co-discovery think-aloud
- Empirical methods and techniques for generating and collecting quantitative UX data
 - Task performance metrics
 - Questionnaires
- Methods and techniques for collecting data about emotional impact and meaningfulness
- Procedures for empirical data collection
 - Preliminaries with participants
 - Session protocol
 - Data collection
- Rapid empirical methods for generating and collecting qualitative UX evaluation data
- Specialized UX evaluation methods

22.1 INTRODUCTION

22.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 22.1). So far in Part 5, we have discussed goals, metrics, and targets for UX evaluation (see [Chapter 20](#)) and preparation to do empirical UX evaluation (see [Chapter 21](#)). This chapter is about doing empirical UX evaluation.

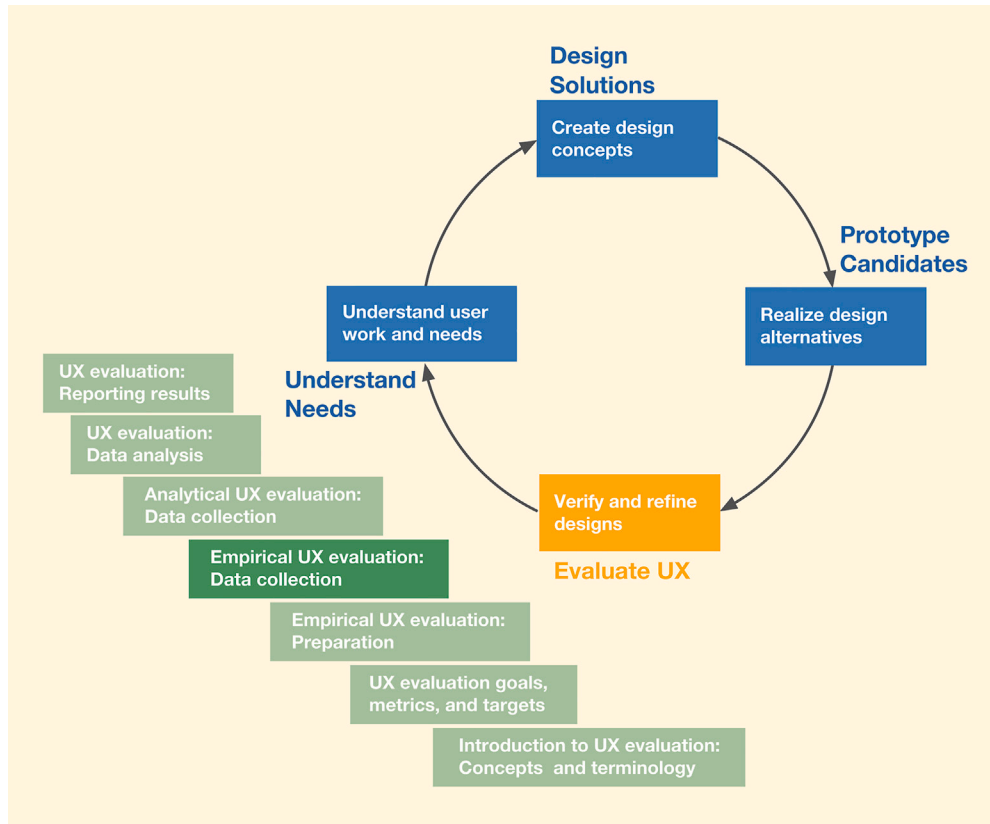


Fig. 22.1

You are here in the chapter on data collection within the evaluate UX activity in the context of the overall Wheel UX lifecycle.

22.1.2 Introduction to Empirical UX Evaluation Methods

22.1.2.1 What qualifies as an empirical method?

There are many definitions of empirical methods, but most have in common the following basic requirements:

- They are first-hand or direct, which means they yield data directly observed and/or measured.
- They are experiential, which means they derive knowledge from actual experience (e.g., of using the system) rather than from theory.

Because, in practice, empirical evaluation means involving users and usage, empirical data are usually observed and/or measured while real user participants perform representative or real tasks. Therefore tasks, including benchmark tasks (see [Section 20.5.1](#)), can be used as empirical measuring instruments for several kinds of UX evaluation data. The same is true for

free-form exercising of the system and exploring it without a specific task in mind, but still experiencing usage.

In addition, user questionnaires and surveys qualify as empirical data collection instruments because the data recorded are first-hand and usually based on usage.

22.1.2.2 *Our treatment of empirical UX evaluation*

We describe empirical UX evaluation methods as generic mainstream methods at a medium level of rigor and a broad-scope view as a foundation. This is a starting point for UX practice as a basis for adaptation, especially to agile UX practice. In [Section 22.7](#), we feature rapid variation methods that can be derived from this more rigorous starting point.

22.1.2.3 *Ecological validity*

Empirical UX evaluation of real systems can take place in a number of settings. The UX lab is a more or less controlled environment, which is a plus in terms of limiting distractions and having all your tools at hand. However, testing in situ (i.e., taking data in the real work context of a field setting) can be more effective at ensuring realistic task conditions. This is ecological validity (Carter et al., 2008), not to be confused with the ecological level of the UX design pyramid (see [Section 11.2.1](#)). Ecological validity is a term that refers to how well the evaluation setup matches real users and real work context. The choice of user participants (see [Section 21.6](#)) or user proxies ensures the former, while ecological validity ensures the latter. Without ecological validity, the findings do not always translate to the real work context or real users. For example, if you are building a system designed for a busy and noisy work context and you evaluate it in a quiet lab without anyone else present, you will not find all design deficiencies that become apparent when the users are interrupted, distracted, or otherwise impacted by the environment.

Example: The Early A330 Airbus—an Example of the Need for Ecological Validity in Testing

I (RH) personally experienced a real-world example of a product that could have benefited enormously from better ecological validity in its testing. I traveled in an A330 Airbus airplane when that model first came out; our plane was 1 week old. (Advice: For many reasons, do not be the first to travel in a new airplane design.) We were told that a human-centered approach was taken in the A330 Airbus design (UX evaluation of buttons, layout, etc.) of

the passenger controls for the entertainment system. Apparently, though, they did not do enough in situ testing. Your personal viewing screen for movies and other entertainment is on the back of the seat in front of you, considered an advantage over the screens hanging from the ceiling. The control is more or less like a TV remote control, only tethered with a pull-out cord. When not in use, the remote control snaps into a recessed space on the seat arm rest. Cool, eh?

Well, as you might guess, this early design had drawbacks. The LCD screen has nice color and brightness but a low effective viewing angle. Get far off the axis (away from perpendicular to the screen) and you lose all brightness; just before it disappears altogether, you see the picture as a color-negative image. But the screen is right in front of you, so no problem, right? Right, until in a real flight the person in front of you tilts back the seat. Then, you can barely see it. I could tell this was affecting others, too, because I could see many people leaning their heads down into an awkward position just to see the screen. After a period of fatigue, many people gave up, turned it off, and leaned back for comfort. If the display screen was used in UX evaluation, and we have to assume it was, the question of tilting the seat never appears to have entered the discussion probably because the screen was propped up on a stand in front of each participant in the UX lab, not emulating the ecological conditions of real flight.

It does not end here. Once the movie began playing, most people stowed the remote control away in the arm rest. But, of course, what do you also rest on an arm rest? Your arm. In so doing, it was easy to bump a button on the control and make some change in the programming. The design of this clever feature almost always made the movie disappear at a crucial point in the plot.

It still does not end here. After the movie was over (or for some viewers, after they gave up) and wanted to sleep, a bump of the arm on the remote caused the screen to light up brightly, instantly waking the passenger to the wonderful world of entertainment. The flight attendants in the first week with this design had already come up with a creative workaround. They showed everyone how to pull the remote out on its cord and dangle it down out of the way of the arm rest. Soon, and this is the UX-gospel truth, almost everyone in the plane had a dangling remote control swinging gracefully in the aisle like so many synchronized reeds in the breeze as the plane moved about on its course.

22.2 EMPIRICAL DATA COLLECTION METHODS: QUANTITATIVE

22.2.1 Objective Data Collection (Task Based)

The most popular quantitative data collection techniques involve measurements of user performance of benchmark tasks (see [Section 20.5.1](#)), possibly at the same time evaluators collect qualitative data. A benchmark task is a task devised for a participant to perform during UX evaluation so that UX measures, such as time on task and error rates, can be obtained and compared to baseline values across the performances of multiple participants.

22.2.1.1 *Timing task performance*

By far the simplest way to measure time on task is by manually using a stopwatch. It is really the only sensible way to measure time with low-fidelity prototypes, such as click-through wireframe prototypes. For the rare times when precise timing measurements of events are required, it is possible to embed software timers to instrument the software internally.

22.2.1.2 *Counting user errors*

The simplest way to count user errors during task performance, especially with low-fidelity prototypes, is to use a manual event counter, such as a handheld clicker for counting people coming through a gate for an event.

The key to counting errors correctly is in knowing what constitutes an error. A user error is usually considered to have occurred when the participant takes any action that does not lead to progress in performing the desired task within the boundaries of the intended design (and not, for example, due to gaps in a prototype's functionality).

22.2.1.3 *A slip is not a user error*

We do not usually include errors called “slips” (Norman, (1990, p. 105). These are errors that users make by accident when, in fact, they know better. For example, the user knows the right button to click but clicks the wrong one, perhaps through a slip of the hand or being too hasty. Also, we do not usually include typing errors, unless their cause could somehow be traced to a problem in the design, or the application is about typing.

22.2.2 Subjective Data Collection (Questionnaires)

Do questionnaires provide empirical data? The short answer is yes; questionnaire data are generally considered to be empirical. But it really does not matter in UX practice because questionnaires are just a data collection method, so the data will be analyzed and reported the same way, regardless. However, we can say that most questionnaires are first-hand (direct) and connected to usage (experiential), so they qualify as empirical.

Advantages of questionnaires. The primary kind of quantitative/subjective data is associated with user surveys and questionnaires to gather data about how users view the design. Questionnaires are an efficient means for getting user feedback on specific targeted aspects of the user experience, including perceived usability, usefulness, and emotional impact.

Often used to supplement an evaluation session with a user participant, questionnaires are simple to use for both evaluator and participant. A questionnaire is a fast, easy, and inexpensive way to collect subjective UX data. Post-session questionnaires can be used to supplement what you have discovered objectively in lab-based data collection sessions. Or a questionnaire can be used as the primary UX data collection instrument when used as an evaluation method on its own.

Questionnaires with good track records, such as the Questionnaire for User Interface Satisfaction (QUIS) (Chin, Diehl, & Norman, 1988), the System Usability Scale (SUS) (Brooke, 1996), or the Usefulness, Satisfaction, and Ease of Use form (Lund, 2001), are easy and inexpensive to use. If you lack the time for making objective performance measurements, the AttrakDiff (Hassenzahl, Burmester, & Koller, 2003) questionnaire may be a good choice because it is designed to address both pragmatic (usability and usefulness) and emotional impact issues.

Semantic differential scales. A semantic differential scale, or Likert (1932) scale, is a mechanism useful for producing quantitative data from possibilities chosen from a questionnaire. A semantic differential scale has a range of semantic values describing an attribute, each value on the scale representing a different level of that attribute. The most extreme value in each direction on the scale is called an anchor. The scale is then divided, usually in equal divisions, with points between the anchors that divide up the difference between the meanings of the two anchors.

For example, consider the following statement for which we wish to get an assessment of agreement by the user: “The checkout process on this website was easy to use.” A corresponding semantic differential scale

may provide these labels: strongly agree, agree, neutral, disagree, and strongly disagree, with the associated numeric values, respectively, of +2, +1, 0, -1, and -2. The user who responds to the questionnaire is guided by the verbal descriptions of the meanings of each choice, but the data yielded are the numeric values. Semantic differential questionnaires make for a product-independent method that can yield reliable quantitative subjective data.

22.2.2.1 Questionnaire for User Interface Satisfaction

QUIS was developed at the University of Maryland (Chin, Diehl, & Norman, 1988). A survey by Stanton et al. (2013) stated that QUIS is one of the earliest available questionnaires for evaluating user satisfaction. It was the most extensive and thoroughly validated questionnaire at the time of its development for determining subjective interaction design usability. It is not clear whether QUIS is still being maintained. A case study illustrated the use of QUIS in evaluating a vending machine (Naeini & Mostowfi, 2015). Reviews of this tool can be found in (Hinderks et al., 2019).

QUIS is organized around such general categories as screen, terminology and system information, learning, and system capabilities, where screen can also refer to a webpage. Within each of these general categories are sets of questions about detailed features, with Likert scales from which a participant chooses a rating. It also elicits some demographic information as well as general user comments about the interaction design being evaluated. Many practitioners supplement QUIS with some of their own questions that are specific to the interaction design being evaluated.

In [Table 22.1](#), we show a sample excerpted and adapted from QUIS with fairly general applicability at least to desktop applications. The columns represent the UX attribute being evaluated, and the semantic anchors for the question.

Modifying QUIS. We have found an adaptation of QUIS to work well. In this adaptation, we reduce the granularity of the scale from 12 choices (0–10 and NA) to 6 (-2, -1, 0, 1, 2, and NA) for each question, reducing the number of choices the participant must face. We thought a midscale value of zero was an appropriately neutral value; negative scale values corresponded to negative user opinions, and positive scale values corresponded to positive user opinions.

Table 22.1

An excerpt adapted from the Questionnaire for User Interface Satisfaction. Adapted from Chin, Diehl, and Norman (1988), used courtesy of the Association for Computing Machinery.

1. Terminology relates to task domain	distantly—closely
2. Instructions describing tasks	confusing—clear
3. Instructions are consistent	never—always
4. Operations relate to tasks	distantly—closely
5. Informative feedback	never—always
6. Display layouts simplify tasks	never—always
7. Sequence of displays	confusing—clear
8. Error messages are helpful	never—always
9. Error correction	confusing—clear
10. Learning the operation	difficult—easy
11. Human memory limitations	overwhelmed—are respected
12. Exploration of features	discouraged—encouraged
13. Overall reactions	terrible—wonderful
Overall reactions	frustrating—satisfying
Overall reactions	uninteresting—interesting
Overall reactions	dull—stimulating
Overall reactions	difficult—easy

22.2.2.2 System Usability Scale

The SUS was developed by Brooke (1996) while at Digital Equipment Corporation in the United Kingdom. The SUS questionnaire contains 10 questions. As an interesting variation from the usual questionnaire, the SUS alternates positively worded questions with negatively worded questions to prevent quick answers without the responder really considering the questions.

The questions are presented as simple declarative statements, each with a 5-point Likert scale anchored with “strongly disagree” and “strongly agree” and with values of 1 through 5. These 10 statements are as follows:

- I think that I would like to use this system frequently.
- I found the system unnecessarily complex.

- I thought the system was easy to use.
- I think that I would need the support of a technical person to be able to use this system.
- I found the various functions in this system were well integrated.
- I thought there was too much inconsistency in this system.
- I would imagine that most people would learn to use this system very quickly.
- I found the system very cumbersome to use.
- I felt very confident using the system.
- I needed to learn a lot of things before I could get going with this system.

The 10 items in the SUS were selected from a list of 50 possibilities and were chosen for their perceived discriminating power.

The bottom line for the SUS is that it is robust, extensively used, widely adapted, and in the public domain. It has been a very popular questionnaire for complementing objective UX data because it can be applied at any stage in the UX lifecycle and is intended for practical use in an industry context. The SUS is technology independent; can be used across a broad range of systems, products, and interaction styles; and is fast and easy for both analyst and participant. The single numeric score is easy to understand by everyone. According to some, SUS was at one time the most highly recommended of all the publicly available questionnaires—and it is (or at least it was) free, whereas QUIS is not.

22.2.2.3 Other questionnaires

Following are other questionnaires that are beyond our scope but may be of interest to some readers.

General-purpose usability questionnaires:

- Computer System Usability Questionnaire, developed by Lewis (1995, 2002) at IBM, is well regarded and available in the public domain.
- Software Usability Measurement Inventory¹ is “a rigorously tested and proven method of measuring software quality from the end user’s point of view.” It is applicable to a range of application types from desktop applications to large domain-complex applications.
- After Scenario Questionnaire, developed by IBM, is available in the public domain (Bangor, Kortum, & Miller, 2008, p. 575).
- Post-Study System Usability Questionnaire, developed by IBM, is also available in the public domain (Bangor, Kortum, & Miller, 2008, p. 575).

¹<https://sumi.usp.ie>

- Single Ease Question (Sauro, 2010; Tedesco & Tullis, 2006), is a “7-Point rating scale to assess how difficult users find a task...Despite its simplicity, we found the SEQ performed about as well or better than more complicated measures of task difficulty.”²
- Subjective Mental Effort Question (SMEQ) (Sauro & Dumas, 2009) is another one-question post-task usability questionnaire. It is reported to have “the potential to provide diagnostic information and be an additional measure of user satisfaction... SMEQ types were easy to learn and quick to execute. The online version of the SMEQ question was highly correlated with other measures and had equal sensitivity to the Likert question type.”

Web evaluation questionnaires:

- Website Analysis and Measurement Inventory³ is a short but very reliable questionnaire that tells you what your visitors think about your website.

Multi-media system evaluation questionnaires:

- Measuring the Usability of Multi-Media Systems is a questionnaire designed for evaluating quality of use of multi-media software products.

Evaluation questionnaires that address emotional impact:

- Lavie and Tractinsky (2004) questionnaire
- Kim and Moon (1998) questionnaire with differential emotions scale

Make up your own questionnaire to evaluate a specific product or system. You can easily find other UX-related questionnaires with an online search; however, we recommend you make up your own questionnaire and ask questions you want answered.

In rigorous scientific studies, modifying a questionnaire can damage its validity. The validity of a questionnaire is a statistical indication of how well the questions address the intent of the questionnaire. Ready-made questionnaires for broad usage are usually created and tested carefully for statistical validity; however, we often need to create our own limited-use questionnaires for a specific system undergoing UX evaluation. To save time and effort, we often do that by modifying an existing questionnaire and not worrying about its validity because validity is rarely a concern in formative evaluation activities of UX practice.

²Jeff Sauro, <https://measuringu.com>, October 2012.

³<http://www.wammi.com/index.html>.

As an example of adapting a data collection technique, you can make up a questionnaire of your own or you can modify an existing questionnaire for your own use by:

- Choosing a sub-set of the questions
- Changing the wording in some of the questions
- Adding questions of your own to address specific areas of concern
- Using different scale values

On any questionnaire that does not already have its scale values centered on zero, you may consider making the scale $-2, -1, 0, 1, 2$, for example, to center it on the neutral value of zero. If the existing scale has an odd number of rating points, you can change it to an even number to force respondents to choose one side or the other of a middle value, but that is not essential here.

Consider adding free-form open questions. One of the downsides of any questionnaire based only on semantic differential scales is that it does not allow the participant to give indications of why any rating is given, which is important for understanding what design features work and which ones do not and how to improve designs. Therefore we recommend you consider adding, where appropriate, a free-form open question (i.e., a short essay question) to gather qualitative data, labeled “If notable, please describe why you gave the above rating.” Open-ended questions can also allow users, user proxies, and others to express opinions about system quality and problems to be solved. This is a case of collecting empirical qualitative data (Section 22.3).

Customer satisfaction (CSAT) studies. Questionnaires or surveys designed to measure satisfaction with a system, product, or service are given to users. One popular type of survey is the CSAT (Allen, Allen, & Rao, 2000), a customer satisfaction survey.

A CSAT study uses a survey typically asking about overall satisfaction (on a Likert scale) with a product or feature and open-ended questions on why they gave that rating, such as problems, frustrations, and gaps in functionality. These can go as far as targeting specific user journeys. For example, a specific question may be: “Doing activity x using the product takes: more time/effort than anticipated, as much time as anticipated, or less time than anticipated.” If the sample size is large enough and formal statistical processes observed, the findings will be statistically significant, meaning claims can be made outside the UX team. For example, the UX team can take the results to management to convince them that there is, or is not, a problem.

22.3 EMPIRICAL DATA COLLECTION METHODS: QUALITATIVE

The defining formative UX evaluation goal is to identify UX problems and their causes so the design can be improved. As such, qualitative data collection is paramount in formative UX evaluation. Among the most popular qualitative data collection techniques used in task-based empirical evaluation with participants are observation and recording of critical incidents, use of the think-aloud technique, and co-discovery. We describe these techniques in the following sections.

22.3.1 Critical Incident Technique for Finding UX Problems

A critical incident is a usage event that reveals something significant about the quality of the user experience. Critical incidents are usually observed by the evaluator, perhaps with help from the participant, usually during task performance.

Although a critical incident can be positive, an indicator of something good in the user experience, for the most part negative critical incidents are what we focus on in UX practice. Therefore, despite many variations in techniques for gathering and analyzing critical incidents, researchers and practitioners agree on the basics, which we now posit as the definition. A critical incident is an event occurring within usage that reveals a barrier, problem, or difficulty encountered by the user, or simply something the user did not like (Castillo & Hartson, 2000; del Galdo et al., 1986). Problems thus identified are traced to their causes in the UX design and put on a list to be fixed in subsequent iterations. Critical incident identification is arguably the workhorse of empirical qualitative data collection techniques.

Critical incident techniques began long ago in human factors. The origins of the critical incident technique can be traced back at least to studies performed in the Aviation Psychology Program of the US Army Air Forces in World War II to analyze and classify pilot error experiences in reading and interpreting aircraft instruments. The technique was first formally codified by the work of Fitts and Jones (1947). From this, Flanagan (1954) synthesized the landmark critical incident technique.

Who identifies critical incidents? In the original work by Fitts and Jones (1947), the user (an airplane pilot) was the one who reported critical incidents after task performance was completed. Later, Flanagan (1954) used trained observers (evaluators) to collect critical incident information while observing users performing tasks.

Del Galdo et al. (1986) involved users in identifying their own critical incidents, reporting during task performance. The technique was also used as a self-reporting mechanism by Castillo and Hartson (2000) and Hartson and Castillo (2000; 1998) as the basis for remote system or product usability evaluation. Further, Dzida, Wiethoff, & Arnold (1993) and Koenemann-Belliveau et al. (1994) adopted the stance that identifying critical incidents during task performance can be an individual process by either the user or an evaluator or a mutual process between the user and an evaluator.

Critical incident data collection. The way you collect and document critical incident data as you go will have much to say about the accuracy and efficiency of your subsequent data analysis. Write concise but detailed critical incident and UX problem descriptions as clearly, precisely, and completely as you can in real time. Terse notes will be more difficult to interpret later.

The best kind of critical incident data are:

- Detailed
- Observed during usage
- Captured immediately
- Associated closely with specific task performance

Critical incident data are perishable. The key to UX problem data is in the details. The biggest reason why empirical UX evaluation is effective is that it allows the capture of detailed usage-related data as it occurs. These detailed data are perishable and likely to be lost if not captured immediately and precisely as they arise during usage.

There are several kinds of retrospective UX problem data capture, including:

- Depending on someone to remember the details in an interview or survey after the session
- Alpha and beta testing, where users are sent software to try out and comment on how well it worked

Retrospective data capture can be used to identify major UX problems, but details of these data are likely to be lost in retrospection.

Sometimes critical incident data are subtle. Experienced UX evaluators will know how to see critical incidents in subtle user behavior—a user hesitation, a participant comment in passing, a head shaking, a slight shrugging of the shoulders, or drumming of fingers on the table. You, the UX evaluator, should

request clarification to determine if any of these subtle observations is a symptom of a UX problem.

What is in critical incident data? Critical incident data about a UX problem should contain as much detail as possible, including:

- User's general activity or task
- Objects or artifacts involved
- The specific user intention and action that led immediately to the critical incident
- Expectations of the user about what the system was supposed to do when the critical incident occurred
- What happened instead
- As much as possible about the mental and emotional states of the user
- Indication of whether the user could recover from the critical incident and, if so, a description of how the user did so
- Additional comments or suggested solutions to the problem

Avoid video recording most of the time. Video recordings of UX critical incidents and think-aloud data collection during UX evaluation sessions were popular in the past for two reasons: (1) to capture every detail, and (2) to provide video evidence of UX problems and the need for redesign solutions. Two things have changed, however, making video recording a less necessary and even undesirable evaluation technique: (1) The faster-paced agile UX process does not allow time for capture or analysis of video, and (2) the credibility of UX teams has risen significantly, obviating the need to prove their worth.

The bottom line is that seldom, if ever, in today's UX practice is it worthwhile to make video recordings of evaluation sessions. Video capture is difficult and expensive, and reviewing the video is time consuming and tedious.

There are a couple of arguments for using video recording in certain cases, however. First, even in the agile UX world, you may want to record video just in case you want to return to some important critical incident and review what happened. Recording video is cheap, and storage is cheap. Second, with the current progress in deep learning and generative artificial intelligence, it is (will be) possible to process video and label it automatically.⁴

Manual note taking for critical incident data collection. Manual note taking is the most useful and efficient critical incident data capture technique. Evaluators take comprehensive, real-time raw critical incident notes with a laptop or a pencil and paper.

⁴Thanks for suggesting this paragraph to Jacek Gwizdka, Information eXperience Lab Director and Associate Professor, School of Information, University of Texas at Austin.

22.3.2 User Think-Aloud Data Collection Techniques

Also called verbal protocol in the early human-factors literature, the think-aloud technique is a qualitative empirical data collection technique in which user participants are encouraged to verbally express their thoughts about the interaction experience as they perform tasks and exercise a UX design. The technique is simple to use for both the UX evaluator and the participant. It is most useful during empirical evaluation of user task performance, but it is also useful when a participant walks through a prototype or helps you with a UX inspection (see [Sections 23.4 and 23.5](#)). Participants divulging their motives, rationale, and perceptions of UX problems reveal otherwise possibly hidden qualitative data about UX problems.

Variations of this simple technique were rooted in psychological and human-factors experimentation well before it was introduced to usability engineering (Lewis, 1982), a precursor to modern-day UX practice.

Advantages of the think-aloud technique. Because the think-aloud technique gets at the user's state of mind precisely at the time of use, giving us access to a precious understanding of the user's expectations, strategies, biases, likes, and dislikes, Nielsen (1993, p. 195) says that “thinking aloud may be the single most valuable usability engineering method.” It is effective in accessing users' intentions, what they are doing or are trying to do, and the reasons why they are doing any particular actions. The think-aloud technique is also effective in assessing emotional impact because emotional impact is felt internally, and the internal thoughts and feelings of the user are exactly what the think-aloud technique accesses.

Following are hints on how to successfully manage participants in the think-aloud technique:

- At the beginning, explain the concept of thinking aloud.
- Explain that you will expect them to talk while they work and think, sharing their thoughts by verbalizing aloud.
- Ask participants to tell you what they are thinking, not describe what they are doing. Sometimes participants resort to giving procedural feedback, such as “I will click here, use this dropdown, ...” Among the thoughts you should encourage participants to express are descriptions of their intentions, what they are doing or are trying to do, and their motivations (the reasons why they are doing any particular actions).
- You may begin with a short exercise or practice session to get warmed up and to get participants acclimated to thinking aloud.
- Encourage them to get past the chatty stage and get down to real engagement and introspection.

- You especially want them to speak out when they get confused, frustrated, or blocked.
- Actively elicit participant thoughts if they are not forthcoming.

Occasionally, you may have to encourage or remind a participant to keep up the flow of thinking aloud, but depending on the individual, thinking aloud usually comes quite naturally and does not take much practice.

22.3.3 Co-discovery Think-Aloud Techniques

You can use two or more participants in a team approach to the think-aloud technique (O'Malley, Draper, & Riley, 1984), an approach that Kennedy (1989) called co-discovery. Although it can seem unnatural and inhibiting to a lone participant to be thinking aloud, essentially talking to oneself, there is more ease in talking in a natural conversation with another person (Wildman, 1995). An individual participant can have trouble remembering to verbalize, but it is natural with a partner. Hackman and Biers (1992) found that using multiple participants, while slightly more expensive, resulted in more time spent in verbalizing, and (more importantly) participant teams produced high-value feedback to designers from multiple viewpoints.

22.4 EMPIRICAL DATA COLLECTION METHODS: EMOTIONAL IMPACT

When your project calls for including emotional impact data in UX evaluation, this section provides methods for doing so.

22.4.1 Direct Observation

Before looking to the more specialized techniques for emotional impact and meaningfulness, your best practical approach is to pick up on indications of emotional impact and meaningfulness during your regular use of critical incident identification and think-aloud techniques.

UX practitioners can obtain emotional impact indicator data through direct observation of participant responses to emotional impact encounters. Usage can be teeming with user behaviors that indicate emotional impact, including verbal and non-verbal languages, gestures, and facial expressions (e.g., grimaces or smiles) (Shih & Liu, 2007, citing Dormann, 2003). For a Usability Test Observation Form, a comprehensive list of verbal and non-verbal behaviors to be noted during observation, see Tullis and Albert (2008, p. 170).

22.4.2 Use Think-Aloud Technique to Evaluate Emotional Impact

In addition to using the think-aloud technique for capturing critical incident information and UX problems, it can also be an excellent window into the mind to access emotional feelings as they occur.

Depending on the nature of the interaction, emotional impact indicators may be infrequent in the flow of task performance user actions, and you may see them mainly as a by-product of your hunt for other UX problem indicators. So, when you encounter an emotional impact indicator during observation in task performance, you certainly should make a note of it. You can also make emotional impact factors the primary focus during the think-aloud technique by making sure participants understand that you want them verbally to share emotional feelings resulting from interaction and usage.

22.4.3 Questionnaires as a Self-Reporting Technique for Collecting Emotional Impact Data

Questionnaires about emotional impact allow you to pose to participants probing questions based on any of the emotional impact factors, such as joy of use, fun, and aesthetics, offering a way for users to express their feelings about this part of the user experience (Westerman, Gardner, & Sutherland, 2006).

22.4.4 AttrakDiff Questionnaire as a Verbal Self-Reporting Technique for Collecting Emotional Impact Data

AttrakDiff (now AttrakDiff2), developed by Hassenzahl, Burmester, and Koller (2003), is an example of a questionnaire based on Likert (semantic differential) scales, especially those developed for getting at user perceptions of emotional impact. AttrakDiff is freely available, short, and easy to administer, and the verbal scale is easy to understand (Hassenzahl, Beu, & Burmester, 2001; Hassenzahl et al., 2000).

22.5 EMPIRICAL DATA COLLECTION METHODS: MEANINGFULNESS

Meaningfulness, or long-term emotional impact, occurs when users invite the product into their lives, giving it a presence in daily activities. To understand the meaningfulness of a product, its adoption into human lifestyles, and how it impacts user lives over long periods of time, you must find ways to study these phenomena over time.

22.5.1 Meaningfulness Evaluation Requires Long-Term Studies

As users experience a product over time, they build perceptions and judgment through exploration and learning as usage expands and emerges (Thomas & Macredie, 2002). Thus meaningfulness is not about tasks but about human activities and presence in human lives. So, naturally, meaningfulness must be studied longitudinally and not just in the snapshots of usage that you may be used to observing in other kinds of UX evaluation. By the time a long-term meaningfulness study is completed, it ends up being a case study. The length of these studies does not necessarily mean large amounts of person-hours, but it can mean significant calendar time. Therefore the technique will not fit with an agile method or any other approach based on a short turnaround time.

It is clear that methods for studying and evaluating the meaningfulness aspects of usage must be situated in real activities of users to encounter a broad range of user experiences occurring naturally. This means that you cannot simply schedule a session, bring in user participants, have them perform, and take your data. Rather, this deeper importance of context usually means collecting data in the field rather than in the lab.

22.5.2 Importance of Self-Reporting

The best raw meaningfulness data would come from constant attention to the user and usage, but it is not possible to live with a participant 24/7 and be in all the places that a busy life takes a participant. The only ones who can be there at all times and places where usage occurs are the participants themselves. Therefore most of the collection techniques for long-term meaningfulness data are self-reporting techniques.

22.5.3 Diary-Based Self-Reporting by Users

We encourage you to improvise a self-reporting technique yourself, but you should definitely consider a diary-based technique in which each participant maintains a diary to document problems, experiences, and occurrences of meaningfulness within usage. Diaries can be kept via paper and pencil notes, online reports, cell-phone messages, or voice recorders. Although diaries are an effective and efficient technique for meaningfulness data collection, analysis of the data can take time and effort.

For a diary-based technique to be effective, participants must be primed in advance:

- Give your users a list of the kinds of things to report, including problems, experiences, and occurrences of meaningfulness within long-term usage.
- Give them some practice exercises in identifying relevant situations and reporting on them.

- Get them to internalize the need to post a report whenever they confront a usage problem, use a new feature, or encounter anything interesting or fun within usage.
- Prompt them periodically with reminders to ensure the participants remember to report these occurrences.

22.6 PROCEDURES FOR EMPIRICAL DATA COLLECTION SESSIONS

22.6.1 Preliminaries With Participants

22.6.1.1 *Introduce yourself and explain the process and your expectations*

Introduce participants to the setup, whether it is a UX evaluation lab, a conference room with a table, or in situ in a real work environment. Show them what happens behind the scenes. Openly declare any video recording you will do, which should have also been explained in the consent form. Make participants feel that they are partners in this endeavor.

Do your best to relieve anxiety and satisfy curiosity. Be sure that your participants have all their questions about the process answered before you proceed into evaluation. Make it very clear that they are helping you evaluate the UX design and that you are not evaluating them.

22.6.2 Prepare Yourself for Evaluating With a Prototype

Almost everything we say here applies to low-fidelity paper prototypes through high-fidelity prototypes running on a laptop or other device. If your wireframe prototype deck is printed on paper, lay it out, and have it ready to go. [Fig. 22.2](#)



Fig. 22.2

Typical setup with participant at the end of a table for evaluation with the prototype deck printed on paper.

shows the participant at the end of the table. The prototype executor is at the left, to operate the wireframe prototype deck. The person at the right is the facilitator and critical incident note taker.

If your wireframe prototype deck is on a laptop, have it ready to operate (by you or by the user participant). We describe the process in terms of a paper wireframe prototype deck, but it works the same way on a laptop.

Before each participant enters, the executor should boot up the prototype by putting the initial screen (e.g., wireframe or webpage) on the table. Arrange everything necessary for running the prototype (e.g., the deck of wireframes).

Have the whole evaluation team ready to assume their roles in the session, including:

- Evaluation facilitator, to interact with participants and keep the session moving
- Prototype executor, to move through the wireframes in response to user actions (a person who knows the design well)
- User performance timer, to time participants performing tasks and/or count errors (to collect quantitative data); the timer may want to practice with a stopwatch before getting into a real session
- Critical incident note takers (for spotting and recording critical incidents and UX problems)

22.6.3 Data Collection Session

22.6.3.1 *The session begins*

If you are using benchmark tasks, after the preliminaries and when you are both ready to begin, have the participant read the first benchmark or other task description and ask if there are any questions. If you are taking timing data, do not include the benchmark task reading time as part of the task.

Now is the time to administer the benchmark tasks and do critical incident identification, use the user think-aloud technique, and apply what you have learned here about the data collection techniques. Once the evaluation session is underway, interesting things can happen quickly. Data you need to collect may start arriving in a flood. It can be overwhelming, but by being prepared you can make it easy and fun.

22.6.3.2 *The facilitator interacts with participants*

The facilitator is responsible for ensuring that the session runs smoothly and efficiently. It is generally the job of the facilitator to listen and not talk. At key

junctures, however, you may gather important data if it does not interfere with task timing or if you are focusing on qualitative data. You can ask brief questions, such as “What are you trying to do?” “What did you expect to happen when you clicked on that icon?” and “What made you think that approach would work?”

If you are focusing on qualitative data, then you may also ask leading questions, such as “How would you like to perform that task?” or “What would make that icon easier to recognize?” If you are using the think-aloud technique for qualitative data gathering, then encourage the participant by prompting occasionally: “Remember to tell us what you are thinking as you go.”

22.6.3.3 *The facilitator keeps the participants at ease*

If participants show signs of stress or fatigue, then give them a break. If participants become visibly flustered, frustrated, zoned out, or blame themselves continually for problems in task performance, they may be suffering from stress, and you should intervene. Let them leave the participant room, walk around, and/or have some refreshments. Do not be too strict about the session schedule.

Take a short break and reassure and calm them. If participants become so discouraged that they want to quit the entire session, there is little you can or should do but thank them, pay them, and let them go.

Exercise 22.1 Empirical UX Evaluation for Your Product or System

This exercise assumes some specifics, such as using the think-aloud data collection technique. You can easily adapt it to your own UX plan and method (see [Section 5.6.1.2](#)).

Goal: To practice empirical data collection for a very simple formative UX evaluation using a wireframe prototype deck.

Activities: This is perhaps the most fun and most rewarding of all the exercises when you finally get to see some users in action with your UX design.

New team formation: This is described in terms of multiple teams in a classroom setting. For other setups, make appropriate adjustments.

- After all the teams are gathered and sitting around a table, make the switch of participants with another team. You send the two people in the participant role from your team to another team. Curb the potential confusion here by doing the swap in an orderly circular fashion among the teams.

- You will now have new participants from a different team who are unfamiliar with your design. These new participants are now permanently on your team for the rest of these exercises, including data collection, analysis, and reporting. As an alternative, if you do not have multiple teams, try recruiting a couple of coworkers or friends as participants.
- Sitting together in your newly formed teams, get out your UX target table form, your benchmark task descriptions, and your questionnaires.
- Dismiss your two participants (the new team members you just got) to the hallway or other waiting area.

Data collection:

- Boot up your prototype.
- Greet your first participant in the waiting area and explain the evaluation session.
- Have this first participant perform your first benchmark task for your UX targets. Have the participant read the first benchmark task aloud.
- Ask the participant to perform that task while thinking aloud.
- The executor moves prototype parts in response to participant actions.
- The facilitator directs the session and keeps it moving.
- Timer(s) write(s) down or enter(s) timing and error count data as indicated in UX targets as the user performs the task (do not count participant's reading aloud of task in task timing).
- Everyone else available should be used to take notes on critical incidents and UX problems.
- Remember the rules about not coaching or anticipating user actions, and the computer (prototype executor) may not speak!
- Have the participant read the second task aloud and ask any questions that might exist.
- Perform it.
- Be ready to collect data according to your UX targets.
- Have this first participant perform your second benchmark task while thinking aloud.

How much data should you collect? You need to collect a dozen or more critical incidents in this overall exercise (i.e., from both participants doing both benchmark tasks). If you do not get at least a half dozen from each participant, continue with that participant doing exploratory use of your prototype until you get enough critical incidents. For example, have them browse through each screen, looking at each object (button, menu,

etc.), commenting on and giving their opinion about the quality of the user experience relating to various features.

Have this participant complete your questionnaire, then give the participant the reward.

Keep your first participant as a new member of your team to help with observations.

Bring in the second participant and perform the same session again.

Deliverables: All your data.

Schedule: Complete by end of class (about 1 hour, if you are efficient).

22.7 RAPID EMPIRICAL UX EVALUATION METHODS

22.7.1 Introduction to Rapid UX Evaluation Methods

So far in this chapter, we have been talking about mainstream UX evaluation methods of medium to relatively high rigor. Such methods, however, can be time consuming and expensive. So, rapid evaluation methods have evolved that are specifically designed with shortcuts to speed them up, to retain as much of the rigor as possible, while greatly increasing efficiency and reducing cost, which are essential for agile UX.

Rapid versions of empirical evaluation methods, which we call quasi-empirical methods ([Section 22.7.2](#)), are mostly abbreviated versions of the regular methods achieved by foregoing certain steps and abridging other steps. Beyond this, most other rapid evaluation methods used in UX practice are analytical methods (e.g., UX inspection methods), which we will visit in [Chapter 23](#).

22.7.2 Quasi-empirical UX Evaluation

Quasi-empirical UX evaluation methods are hybrid approaches arising when UX professionals develop their own empirical UX evaluation methods with shortcuts. They are empirical because they involve data collection (of at least qualitative data) using participants or participant surrogates. They are quasi because they are abbreviated, informal, and flexible with respect to process and protocol, and the UX evaluator can play a significant analytic role. There also can be a significant analytic component.

Quasi-empirical UX evaluation methods:

- Are still mostly empirical, using volunteer participants (or an evaluator emulating a user)
- Are defined by the freedom given to the practitioner to innovate—to make it up during the process while being flexible about goals and approaches

- Are very informal with respect to protocol—evaluators are encouraged to interrupt and intervene at opportune moments to elicit more thinking aloud and to ask for explanations and specifics
- Do not usually involve quantitative data
- Are often punctuated with impromptu changes of pace, changes of direction, and changes of focus
- Are characterized by jumping on issues as they arise and milking them to get the most information about problems, their effects on users, and potential solutions
- Are not based on pre-defined benchmark tasks, but a session can be task driven, drawing on usage scenarios, essential use cases, step-by-step task interaction models, or other task data or task models
- Are not characterized by large numbers of participants, but are intended for getting the most out of a very few participants
- Can be driven by exploration of features, screens, widgets, or whatever suits

22.7.3 Rapid Iterative Testing and Evaluation UX Evaluation Method

Rapid iterative testing and evaluation (RITE) is an example of a specific rapid UX evaluation method.

22.7.3.1 Introduction

RITE (Medlock et al., 2005; Medlock et al., 2002) is a user-based UX evaluation approach designed to pick low-hanging fruit at relatively low cost.

The defining feature of RITE is fast turnaround—fixing key UX problems as soon as they are identified. Immediately after the UX data are collected, the whole project team, including the participants, analyzes the problems and decides on which changes to make. Changes are then implemented straightaway. If warranted, another immediate iteration of testing and fixing may ensue. This fast turnaround of qualitative evaluation results and problem fixing makes RITE one of the most agile empirical methods.

Problem reporting occurs while the team is still there, so they are already informed and immersed in the process. Because changes are included in all versions evaluated after that point, further testing can determine the effectiveness of the changes—that is, whether the problem is, in fact, fixed and whether the fix introduces any new problems. Fixing a problem immediately also gives access to any aspects of the product that could not be tested earlier because they were obscured by that problem.

In his inimitable wixonian wisdom, our friend Dennis reminds us, “In practice, the goal is to produce, in the quickest time, a successful product that meets specifications with the fewest resources while minimizing risk” (Wixon, 2003).

22.7.3.2 *How to do it: The RITE UX evaluation method*

This description of the RITE UX evaluation method is based mainly on Medlock et al. (2005; 2002).

The project team begins by selecting a UX practitioner, whom we call the facilitator, to direct the testing session. The UX facilitator and the team prepare by:

- Identifying the characteristics needed in participants
- Deciding on which tasks they will have the participants perform
- Agreeing on critical tasks, the set of tasks that every user must be able to perform
- Constructing a test script based on those tasks
- Deciding how to collect qualitative user behavior data
- Recruiting participants (see [Section 21.6](#)) and scheduling them to come into the lab

The UX facilitator and the team conduct the evaluation session with one to three participants, one at a time, by:

- Gathering the entire project team and any other relevant project stakeholders, either in the observation room of a UX lab or around a table in a conference room
- Bringing in the participant playing the role of user
- Introducing everyone and setting the stage, explaining the process and expected outcomes
- Making sure that everyone knows the participant is helping evaluate the system, and the team is not in any way evaluating the participant
- Having the participant perform a small number of selected tasks while all project stakeholders observe silently
- Having the participants think aloud as they work
- Working together with the participants to find UX problems and ways the design should be improved
- Taking thorough notes on problem indicators, such as task blocking and user errors
- Focusing session notes on finding usability problems and noting their severity

The UX facilitator and other UX practitioners:

- Identify from session notes the main UX problems observed and their causes in the design
- Give everyone on the team the list of UX problems and causes

The UX practitioner and the team, including the participants, address problems:

- Identifying problems with obvious causes and obvious solutions, such as those involving wording or labeling, to be fixed first
- Determining which other problems can also reasonably be fixed
- Determining which problems need more discussion
- Determining which problems require more data (from more participants) to be sure they are real problems
- Sorting out which problems they cannot afford to fix right now
- Deciding on feasible solutions for the problems to be addressed
- Implementing fixes for problems with obvious causes and obvious solutions
- Starting to implement other fixes and bringing them into the current prototype as soon as feasible

The UX practitioner and the team immediately conduct follow-up evaluation by:

- Bringing in new participants
- Having them perform the tasks associated with the fixed problems, using the updated design
- Working with the participants to see if the fixes worked and to be sure the fixes did not introduce any new UX problems

The entire process just described is repeated until you run out of resources or the team decides it is done (all major problems found and addressed).

22.7.3.3 Variations in RITE data collection

The flexibility of RITE allows consideration of alternative data collection techniques. For example, instead of testing with user participants, the team could employ a UX inspection method, heuristic evaluation, or other techniques for data collection while retaining the fast analysis and fixing parts of the cycle.

22.8 SPECIALIZED UX EVALUATION METHODS

In addition to the standard UX evaluation methods of the previous sections, there are a number of specialized methods. We briefly describe a few here as a matter of interest.

22.8.1 Post-deployment Field Testing: Alpha and Beta Testing and Field Surveys

Alpha and beta testing are useful post-deployment evaluation methods. After almost all development is complete, manufacturers of software applications sometimes send out alpha and beta (pre-release) versions of the application software to select users, experts, customers, and professional reviewers as a preview. In exchange for the early preview, users try it out and give feedback on the experience. Often little or no guidance is given for the review process beyond survey questions, such as “Tell us what you think is good and bad and what needs fixing” and “What additional features would you like to see?”

22.8.2 Remote UX Evaluation

Remote UX evaluation, moderated or unmoderated, has recently become much more frequently used. In many large companies whose business is mostly or all web based, empirical UX evaluation is almost all remote. This has happened for several reasons:

- Users and UX team members are geographically distributed.
- The pandemic forced people to work remotely.
- Remote evaluation is less costly.

Typically, the evaluation is moderated, the only real difference being that they are not colocated. The interaction occurs over video-conferencing tools (e.g., Zoom), and screen-sharing features are used where users launch the prototype on their system and share their screens with the evaluation team.

Remote UX evaluation methods are not a new idea (Dray & Siegel, 2004; Hartson & Castillo, 1998). Methods include:

- Simulating lab-based UX testing using the internet as a long extension cord to the user (e.g., UserVue by TechSmith)
- Online surveys for getting after-the-fact feedback
- Software instrumentation of clickstream and usage event information
- Software plug-ins to capture user self-reporting of UX issues (Hartson & Castillo, 1998)

As always, the best feedback for design improvement is feedback deriving from Carter's (2007) "inquiry within experience," or formative data collected concurrently with usage rather than retrospective recollection.

22.8.3 Automatic UX Evaluation

Lab-based and UX inspection methods are labor intensive and therefore limited in scope (small number of users exercising small portions of large systems). Large and complex systems with large numbers of users, however, offer the potential for a vast volume of usage data. Think of observing 100,000 users using Microsoft Word. The result is a massive amount of data about keystrokes, clickstreams, and pause/idle times. Data collected through such instrumentation can be analyzed to evaluate metrics, such as time to complete tasks or points in a task sequence where users abandon a transaction. Automatic methods have been devised to take advantage of this boundless pool of data, collecting and analyzing usage data without need for UX specialists to deal with each individual action. This type of data could be useful in identifying potential problems with a design. For example, for systems with high usage (e.g., the checkout flow of a popular e-commerce website), if a small portion of the users who start the flow complete it, and most abandon it at a particular step (as identified by the instrumentation), it is reasonable to assume there is a problem at that step, and it should be investigated with a more targeted evaluation.

Unfortunately, all data are at the low level of user actions, without any information about tasks, user intentions, or cognitive processes. It is often difficult to identify when the user is having a UX problem in the midst of that torrent of user action data. Basing redesign on click counts and low-level user navigation within a large software application could well lead to low-level optimization of a system with a bad high-level design.

22.8.4 A/B Comparison Testing in UX

A/B testing⁵ is an effective and economical way to assess the impact of design variations on various criteria, such as user performance, by determining which version of a feature in your system is best. This is useful, for example, in determining whether changes you make in a product or system actually yield an improvement in system and/or user performance.

A/B testing is quantitative, empirical, and statistical and can involve performance measures or post-use subjective data, such as participant

⁵https://en.wikipedia.org/wiki/A/B_testing

opinions expressed via polls or surveys. Amazon and Google, among others, are massive users of A/B testing, having many tests running at any instant in time.

As used in UX, A/B testing employs a formal randomized (with respect to participant group assignment) experimentation process to compare the performance of two (or more) instances of UX design elements according to some criterion. Typically, two versions of a design element, such as a feature within a website, are shown to, or used by, different sets of website visitors to determine which is the best design. The one that has the most impact with respect to business goals is the winner. Implementing the changes found in the winning version helps optimize your website and increases business return on investment in the process.

A/B testing requires that the system being evaluated be instrumented (e.g., by embedding measuring and tracking probes to capture fine-grained user actions and system responses in a log that can be analyzed). Large websites with high volumes of user traffic are an excellent source of this kind of data because of the ease with which they can be instrumented and the large amounts of data that can be generated. The large amounts of data combined with rigorous statistical methods put A/B testing in the formal summative category.

There are many organizations with systems that support large numbers of users on a daily basis (thousands or millions of visits [uses] per day). These systems are often expensive to create and maintain, and given their large usage numbers, it pays to make them as effective as possible. Because of the high volume of users, relatively small design changes (e.g., to fix a known issue or improve a feature) can make large changes in system effectiveness with a correspondingly large business impact. To mitigate risks, it is therefore important to measure the impact of each change that is introduced.

In systems such as these, key user activities (sequences of user tasks, such as searching for and buying a product on an e-commerce website) are deconstructed and described at a granular or atomic level. Various success criteria for these user activities are defined and could include such measures as time to complete the transaction from landing on the home page, conversion rate (percentage of users who complete a transaction), and subjective satisfaction via inbuilt survey measures. Each such criterion is then instrumented with appropriate hooks in the system code for tracking user clicks or taps on various elements on the screen, time spent scrolling, number of times navigation or other elements were accessed, and number of help or other support resources were accessed.

The data collected from such instrumentation are analyzed and curated to feed product health and business dashboards to assess trends with product usage. When a design change is proposed for the system, it is introduced as a new version complete with instrumentation of all key metrics tracked in the previous version. Typically, a portion of the website's traffic is diverted to the new B version of the design, and data collected from this B condition are then analyzed using statistical methods to assess how it compares to the current A version. If it is found to be better, the portion of traffic coming to the site is slowly increased to the new version before eventually switching over by cutting off the old version. It is not uncommon for large systems to have many sets of A/B tests running at any given time.

How it works. We include an overview of A/B testing here without elaborating in great detail on how to use it. If you work in an organization that uses A/B testing, you will be following the A/B testing data collection and analysis procedures already in place at your organization.

Davis (2022) described how a typical application of A/B testing may be used. As a simple example, suppose the user activity is the task sequence involved in signing up for your organization's newsletters, and the current version of the site is not getting enough users to sign up.

You create a new version (B) of the part of your site where you try to get users to sign up. For example, maybe your original design (A) simply has a button to click if a visitor wants to subscribe. Your new design could have an explicit call to action urging visitors to subscribe and explaining benefits. Point your A/B testing tool to these two versions and tell it exactly what data to collect. For example, for each visitor to either version of this webpage, you want to record whether they clicked to subscribe.

Run a test. The A/B testing tool will automatically and randomly, by one of many ways, siphon off two streams of user visits, directing each to a different version. The tool keeps data for both versions on visitors and conversions (those who signed up). Let the test run until there are sufficient data to make a conclusion. Then have the tool make the statistical data analysis comparing the versions. Decide on the winning design, which will already be implemented and deployed. Switch the entire user traffic over to the winning version.

Data analysis is the strength of software tools for A/B testing. The complexities of rigorous statistical analysis of A/B experiments are handled inside the tool, saving time and labor by your not having to re-invent or re-learn this mathematical process. If your design situation and needs fit, then data analysis can come down almost to a plug-and-play operation.

A/B testing software tools. At the end of the day, A/B testing tools are important because the way you end up doing A/B testing in UX practice is through one of the tools. The tools guide you as to what parameters you need to provide and how to provide them. Most of the tools available exist online, and you can direct them to apply to your website and its traffic.

The number and variety of A/B testing support tools are constantly changing, and we cannot track the evolution in a book. We also cannot offer tool reviews or recommendations.

22.9 SEGUE

In this chapter, we discussed empirical data collection techniques. In the next chapter, we move on to analytical versions.